MILITARY CLASSIFICATION: ARMED FORCES INTERNAL USE ONLY • Confidential Welfare Engineering Guide

APPS Platform - Muzhu Features & Usage Manual

AI Predictive Personnel Stress & Welfare Monitoring System (Tanglish Edition)

Indha official technical document-la namma APPS system start panna naal-la irundhu ippo varaikkum build panna **ellaa features, tools, clinical algorithms, security mechanisms, matrum UI components** patri detailed-ah Tanglish-la vilakkapattulladhu. Indha platform soldier psychological health-ai early-stage-la analyze panni, severe burnout, depression, matrum PTSD issues-ai prevent panna design panna pattu irukku. Ovvoru feature-um epadi work aagudhu, adhai epadi use pannanum, yaaru use pannanum enbadhai keezhe detailed-ah paarkkalaam.

Document Title:	APPS Complete System Features & User Manual (Tanglish)
Target Audience:	Soldiers (Personnel), Welfare Officers, Senior Command Admins, Developers
System Core Engine:	Python Flask, SQLite3, Vercel Blob Persistent Snapshot Engine, CSS3 Flex/Grid
Security Baseline:	PBKDF2 Password Salting, Command PIN Gating, Immutable Activity Audit Trail
Live Production URL:	https://apps-nu-henna.vercel.app (Deployed on Vercel)
Repository:	https://github.com/dharuntech275/APPS- (Dual tracked: main & master)

1. System Overview & 3-Tier Command Architecture

APPS platform armed forces jawan-galukku matrum welfare officers-kku oru reliable digital bridge-ah vela seiyudhu. System-la motham **3 mukkiya roles** irukku:

- **1. Personnel (Soldier / Jawan):** Dhinamum thanga stress, thoongum neram (sleep hours), heart rate, matrum family morale status-ai log panni thanga mental fitness-ai track pannalaam. Instant panic relief-ku 4-7-8 breathing exercise matrum emergency 1-click Crisis SOS call seiyalaam.
- **2. Welfare Officer:** Thanga command unit-la irukkum anaithu jawans-oda health status-ai live-ah monitor pannalaam. High-risk red alert jawans-ai prioritize panni clinical counseling notes add panni suicide prevention actions edukalaam.
- **3. Senior Admin (HQ Command):** Unit-level statistics (High, Medium, Low risk counts), dynamic Command Master PIN update, audit trail activity logs, matrum overall system oversight control seiyalaam.

Mukkiya Kurippu: Starting-la login issue irundhadhu, adhai streamline panni login direct aakkiyom. Pinnar 2FA shield kooda thevai illai enbadhaal UI, database, matrum backend flow-la irundhu clean-ah remove panni friction-free direct login ready panniyom.

2. Authentication, Account Management & Security Features

2.1 Direct Secure Login & Session Management

Enna Feature: User thanga username matrum password enter panni direct-ah thanga dashboard-kku login seiyyum system. 2FA TOTP shield thevai illai endru sonnadhukku apparam, 6-digit authenticator code thevai illamal direct single-step login create panni irukkom.
Epadi Work Aagudhu: User credentials enter pannumpodhu, backend-la SQLite database query panni password verify aagum. Password match aana udane HTTP-only permanent signed session create aagi, user-oda role-kku thagundha dashboard-kku redirect aagum (`/dashboard`, `/officer/personnel`, or `/admin_dashboard`).
Benefit: Field-la network speed kuraivaaga irukkum podhum jawan-gal friction illamal 2 seconds-la login panni daily log submit panna mudiyum.

2.2 Command Authorization Master PIN Security

Enna Feature: Pudhu user sign-up aagumpodhu, yaravathu thavaraaga 'Welfare Officer' alladhu 'Senior Admin' role select panni unauthorized access vaanga koodathu enbadharkaana master defense barrier.
Epadi Work Aagudhu: Signup page-la 'Officer' alladhu 'Admin' role select panninaal, mandatory-ah 'Command Security PIN' kekkum. Indha PIN database-la `SystemConfig` table-la secure-ah store aagi irukkum (Default: `2026`). Senior Admin mattum thaan thanga dashboard-la irundhu indha PIN-ai eppodhu vendumanalum change panna mudiyum. Soldier / Personnel role-kku PIN thevai illai.
Benefit: Unauthorized person yaarum officer account create panna mudiyadhu, command hierarchy 100% strictly maintain aagum.

2.3 Cryptographic PBKDF2 Password Hashing & Weak Password Upgrade

Enna Feature: Security upgrade feature. Pazhaiya plain-text alladhu weak passwords-ai defense-grade PBKDF2-HMAC-SHA256 salting-kku automatic-ah upgrade seiyyum mechanism.
Epadi Work Aagudhu: User login pannumpodhu password weak-aga irundhaalo alladhu unhashed-aga irundhaalo, system udane `/upgrade_password` modal popup kaattum. User 8+ characters strong password set panniyavudan cryptographically hash aagi database-la update aagividum.
Benefit: Military staff accounts brute-force attacks matrum credential theft-la irundhu safe-ah irukkum.

2.4 Self-Service Password Recovery (/forgot_password)

Enna Feature: Soldier password marandhu poitaal, external email alladhu SMS OTP thevai illamal thanga unit verification moolam password-ai reset pannum defense self-service tool.
Epadi Work Aagudhu: Login screen-la 'Forgot Password?' click panni, Registered Name matrum Service Unit (e.g. 15 Rajputana Rifles) submit pannanum. Record match aana udane puthiya password set panna allow pannum.
Benefit: Remote border outpost-la phone network / OTP service illadhavargalukku indha unit-verified recovery migavum help-ah irukku.

2.5 Login Page Streamlining & Cleanup

Enna Feature: Neenga ketta maadhiri login page-la irundha thevai illadha banners, unnecessary test links, matrum clutter-ai remove panni clean executive login card-ah maathiyadhu.
Epadi Work Aagudhu: Clean card layout, direct username/password fields, role-based registration link, matrum emergency hotline reference mattum display aagum.
Benefit: User confuse aagaamal fast-ah login seiyalaam.

3. Personnel (Soldier / Jawan) Features & Detailed Usage

3.1 Daily Mental Health & Wellness Assessment Survey

Enna Feature: Jawan thanga daily psychological and physical health telemetry-ai submit pannum core assessment form. Idhula 4 clinical metrics irukku:
• **1. Sleep Duration (Hours):** 0 mudhal 12 mani neram varai thoongiya alavu.
• **2. Operational Stress Level (1 - 10 Slider):** Interactive slider. Move pannumpodhu Dynamic Qualitative Badges marum: 1-3 (Calm & Focused - Green), 4-5 (Manageable Duty Load - Slate), 6-7 (Elevated Tension - Amber), 8-10 (Critical Combat Stress / High Risk - Red).
• **3. Heart Rate / Pulse (BPM):** 40 mudhal 180 bpm varai physical exertion alavu.
• **4. Social / Family Morale Connection (1 - 5 Stars):** Family and squad support level.
• **Field Operational Notes:** Thangalukku irukkum bayam, thoongamai, alladhu duty shramathai pathi ezhuthalaam.
Epadi Use Pannanum: Personnel dashboard-kku vandhu sliders-ai adjust panni, 'Log Daily Wellness Entry' button-ai click pannanum.
Benefit: Jawan-oda daily mental status continuous-ah track aagum.

3.2 Automated Psychological Triage Scoring Algorithm

Enna Feature: Jawan log submit panna udane, AI clinical heuristic algorithm automatic-ah triage level-ai calculate pannum:
• **High Risk (Urgent Action Required - Red):** Sleep < 5 hours, Stress >= 8, Heart Rate > 100 bpm, alladhu Social Connection == 1.
• **Medium Risk (Moderate Caution - Amber):** Sleep 5-6 hours, Stress 5-7, alladhu Heart Rate 90-100 bpm.
• **Low Risk (Stable Mental State - Green):** Adequate sleep (>= 7h), Stress < 5, normal heart rate.
Benefit: Welfare officers immediate-ah yaara contact pannanum-nu automated-ah theriyum.

3.3 Tactical 4-7-8 Breathing & Relaxation Stabilizer

Enna Feature: Jawan-galukku panic attack, battle stress, alladhu sleeplessness varumpodhu parasympathetic nervous system-ai activate seiyyum breathing tool.
Epadi Work Aagudhu: Dashboard-la '4-7-8 Relax' quick action button alladhu full card irukku. 'Start Session' click panninaal:
1. **Inhale (4s):** Animated circle expand aagum, mooku valiyaaga moochu ulle edukka vendum.
2. **Hold (7s):** Circle steady-ah irukkum, moochai adakki vaikka vendum.
3. **Exhale (8s):** Circle shrink aagi moochai vaai valiyaaga methuvaaga veliye vida vendum.
Dynamic cycle counter (Cycle 1 of 4) matrum animated progress bar display aagum.
Benefit: 2 minutes-kulle heart rate kuraithu instant mental calmness kidaikkum.

3.4 Simulated Smartwatch / BLE Telemetry Sync

Enna Feature: Wearable fitness bands alladhu army smartwatches (BLE) illadha idathil kooda, telemetry sensor-ai simulate panni single click-la realistic sleep and heart rate data-vai fill pannum tool.
Epadi Use Pannanum: Daily survey form-la irukkum 'Sync Smartwatch Data' button-ai click seiyyum podhu, realistic military biometrics (Sleep: 6.8h, Heart Rate: 72 bpm) automatic-ah form fields-la load aagividum.
Benefit: Form fill seiyyum neram kuraivadhaal jawans regular-ah entry poduvargal.

3.5 1-Click Crisis SOS Emergency Beacon

Enna Feature: Soldier extreme mental breakdown, suicidal thoughts, alladhu severe panic situation-la irukkumpodhu 1-click-la emergency help kidaikkum defense beacon.
Epadi Work Aagudhu: Header-lum dashboard banner-lum prominent RED SOS button irukku. Click panninaal:
1. Army Emergency Helpline (**1902**) & National Tele-MANAS (**14416**) direct call modal open aagum.
2. Welfare Officer matrum Admin screen-la urgent red pulsing banner alert dispatch aagum.
Benefit: Life-saving crisis intervention immediate-ah kidaikkum.

3.6 Cumulative Fatigue Forecaster & 7-Day Sleep Debt Engine

Enna Feature: Soldier kadandha 7 naatkalaga evvalavu thookkam izhanthirukirar (sleep debt) enbadhai calculate seiyyum clinical algorithm.
Epadi Work Aagudhu: Baseline military requirement 7 hours daily sleep. Daily sleep deficit-ai kooti **Operational Strain Index (0 - 100%)** calculate seiyudhu. Strain index 65%-kku mela ponaal, 'Severe Sleep Debt Alert - Mandatory Rest Required' endru recommendation kaattum.
Benefit: Prolonged fatigue aala varum weapon handling accidents matrum guard duty lapses-ai thadukkum.

3.7 Gamified Morale Badges & Daily Streak Tracking

Enna Feature: Jawans thodarndhu daily entry pottu thanga health-ai track panna motivate pannum gamification system.
Epadi Work Aagudhu: Consecutive days entry pottaal streak increase aagum (e.g. 5-Day Streak ■). Athil 'Mental Resilience Sentinel', 'Sleep Master', 'Iron Mind' pontra military achievement badges unlock aagum.
Benefit: Mental health entry oru duty-ah feel aagaamal positive habit-ah maarrum.

3.8 Soldier Personal Health Dossier PDF Export (/my/export_pdf)

Enna Feature: Jawan thanga mudiyatha health history, past surveys, triage levels, matrum fatigue index-ai professional medical PDF report-ah download seiyyum feature.
Benefit: Base medical officer-kku physical check-up panna poga podhu indha PDF report-ai katti consultation edukalaam.

4. Welfare Officer Command Suite & Clinical Management

4.1 Personnel Live Monitoring Roster & Triage Filter

Enna Feature: Officer thanga command unit-la irukkum anaithu jawans-oda current psychological triage status-ai live-ah paarkkum central dashboard.
Epadi Work Aagudhu: Personnel list table-la ovvoru jawan-oda Name, Service Unit, Latest Sleep, Stress Level, Heart Rate, Triage Tag (High/Medium/Low), matrum Action link display aagum. Search bar matrum filter moolam High Risk jawans-ai mattum instant-ah isolate panni paarkkalaam.
Benefit: 100+ jawans irundhalum, risk-la irukkum 3 jawan-ai 2 seconds-la identify panna mudiyum.

4.2 Soldier Longitudinal Clinical Dossier

Enna Feature: Oru specific jawan-oda complete past records, date-wise mental trends, sleep debt graph, matrum biometric changes-ai kaattum detailed profile.
Epadi Use Pannanum: Personnel list-la jawan name pakkathula 'View Dossier' click seiyya vendum.
Benefit: Sudden mood drop alladhu prolonged combat fatigue eppo start aanadhu endru accurate-ah track panna mudiyum.

4.3 Confidential Clinical Counseling Notes Engine

Enna Feature: Welfare Officer jawan-kku 1-on-1 counseling kuduthu vittu, thanga clinical findings, action steps, matrum next review date-ai confidential-ah log pannum tool.
Epadi Work Aagudhu: Dossier page-la 'Add Clinical Counseling Note' form irukku. Officer note type (General Counseling, PTSD Intervention, Family Stress) select panni text submit panninaal, athu encrypted SQLite `CounselingNotes` table-la save aagum matrum `ActivityLog`-lum immutable audit entry vizhum.
Benefit: Medically structured record maintain aagum; soldier transfer aanaalum new officer-kku past background theriyum.

4.4 Emergency SOS Alert Banner Intercept

Enna Feature: Jawan SOS button-ai trigger panninaal, officer-oda screen-la top position-la pulsing crimson RED banner display aagum.
Benefit: Officer vera edhavathu page-la irundhaalum, emergency case-ai miss panna maattaargal.

4.5 Unit Welfare Assessment Report PDF Export (/officer/export_report)

Enna Feature: Motha unit-oda triage distribution, high-risk case summary, matrum counseling follow-up details-ai formal military briefing PDF document-ah export seiyyum tool.
Benefit: Commanding Brigade alladhu Battalion Headquarters-kku weekly welfare status report submit panna idhu direct-ah use aagum.

5. Senior Admin Command Headquarters & Operations

5.1 Real-Time Executive Command Analytics Dashboard

Enna Feature: Headquarters Admin-kku armed forces units-oda overall mental readiness-ai bird's-eye view-la kaattum analytics overview.
Epadi Work Aagudhu: Total Registered Personnel count, High Risk Soldiers count (Red Card), Medium Risk count (Yellow Card), and Stable Low Risk count (Green Card) live metric cards-ah display aagum. Unit-wise summary table-la ovvoru regiment/battalion-la evvalavu per risk-la irukkanga endra data varum.
Benefit: Top brass commanders endha unit-la operational stress adhigama irukku-nu therindhu squad rotation alladhu leave sanction seiyalaam.

5.2 Dynamic Command Master PIN Rotation Engine

Enna Feature: Officer and Admin registration security-kku thevaiyaana 4-digit master PIN-ai Senior Admin instant-ah change pannum facility.
Epadi Use Pannanum: Admin dashboard-la 'Update Master PIN' input box-la pudhiya 4 to 6 digit number (e.g. `2027`) enter panni 'Update' click panninaal udane live-ah update aagividum.
Benefit: PIN leak aanaal kooda database alladhu code-ai thodaamal 5 seconds-la live-ah PIN-ai rotate pannividalaam.

5.3 Immutable Operational Audit Trail (ActivityLog)

Enna Feature: System-la nadakkum anaithu critical actions-aiyum timestamp, actor ID, target soldier, matrum action type-udan record pannum defense audit trail.
Epadi Work Aagudhu: Login, counseling note entry, PIN change, triage status transition ellaame append-only table-la save aagum. Yaarume indha logs-ai edit alladhu delete panna mudiyadhu.
Benefit: Military security compliance matrum complete operational accountability urudhi aagum.

5.4 Senior Command Summary PDF Export (/admin/export_report)

Enna Feature: Top-level military executive report ready pannum generator.
Benefit: Monthly readiness briefings-kku presentation-ready formatted report kidaikkum.

6. UI/UX Layout, Localization & Cloud Infrastructure

6.1 Multi-Language Vernacular Switcher (English, ■■■■■, ■■■■■)

Enna Feature: Soldier thanga thaa mozhiyil (Tamil, Hindi, alladhu English) system-ai use seiyyum client-side instant language toggle.
Epadi Work Aagudhu: Header-la dropdown select box irukku. Language choose panninaal zero-latency-la page-la irukkum ellaa text-um instant-ah switch aagum. Indha preference browser-oda `localStorage`-la save aavadhaal, adutha murai open pannumpothum avargalukku thaa mozhiyileye varum.
Benefit: English theriyadha jawans kooda easy-ah survey fill panni mental health assistance pera mudiyum.

6.2 Laptop Executive 2-Column Grid & Mobile Responsive Drawer Layout

Enna Feature: Neenga ketta laptop view and mobile alignment request-kku yetharpola build panna responsive design system.
Epadi Work Aagudhu:
• **Laptop Screens (>= 992px):** Modern 2-column executive grid (`1.15fr 0.85fr`). Left column-la survey card, right column-la fatigue forecaster matrum gamified morale badges pakkam pakkamaaga clean-ah render aagum. Horizontal scroll illamal compact executive presentation kidaikkum.
• **Mobile Screens (< 768px):** Touch-friendly single column vertical flow. Navigation bar hamburger icon moolam slide-out drawer menu-ah maarrum. Quick action buttons (`Crisis SOS` and `4-7-8 Relax`) side-by-side equal height wrap aagi touch panna comfortable-ah irukkum. Page-la horizontal overflow (`overflow-x: hidden`) completely zero aakki irukkom.
Benefit: Soldier smartphone-la use panninaalum, Welfare Officer laptop-la use panninaalum flawless layout kidaikkum.

6.3 Vercel Serverless Architecture & Private Vercel Blob Persistence Engine

Enna Feature: Cloud-la deployment panna podhu, serverless container shut down aanaalum database alindhu pogamal thadukkum zero-data-loss synchronization engine.
Epadi Work Aagudhu: Vercel-la standard container ephemeral (temporary) aagum. Idhai solve panna custom `sync_db_to_blob()` matrum `sync_db_from_blob()` engine build panniyom. Ovvoru murai survey, note, alladhu PIN update pannumpothum, backend database SQLite file snapshot private Vercel Blob store (`apps-data` store ID: `store_eS9r7map43Dx32yX`)-kku instant-ah upload aagum. Pudhiya container wake up aagumpodhu blob store-la irundhu restored database automatic-ah hydrate aagividum.
Benefit: Free cloud hosting-la kooda 100% persistent defense database guarantee kidaikkudhu.

6.4 2FA Shield Removal & Login Streamlining Refactor

Enna Feature: Starting-la build panna Google Authenticator (RFC 6238 TOTP) shield-ai, user request-kku inanga complete-ah remove panniyom.
Enna Maathinom:
1. Desktop header matrum mobile drawer-la irundha '2FA Shield' links anaithum neekkiyom.
2. Admin Dashboard-la irundha '2FA Security Roster Card' matrum reset buttons neekkiyom.
3. Login challenge intercept (`/login/2fa`) remove panni direct single-step login create panniyom.
4. Database-la irundha `totp_enabled` flags reset panni Vercel Blob cloud store-lum sync panniyom.
Benefit: Login process speed adhigamaagi, zero unnecessary verification step-la soldiers access panna mudiyudhu.

7. Comprehensive Feature Matrix & Usage Cheatsheet

Keezhe kanda table-la namma platform-oda anaithu features, athai yar yaaru use pannalaam, matrum adhan primary use case quick reference summary-ah kudukappattulladhu:

Feature Name	Access Role	Enna Purpose & Epadi Use Aagum (Tanglish Summary)	Status
Daily Assessment Survey	Personnel	Sleep (0-12h), Stress slider (1-10), Heart rate, Social connection log panna.	**Active**
Triage Scoring Algorithm	System / All	Stress and sleep data vechu High (Red), Medium (Yellow), Low (Green) triage auto calculate aagum.	**Active**
4-7-8 Breathing Stabilizer	Personnel	Animated circle guide vechu 4s Inhale, 7s Hold, 8s Exhale seithu panic relief pera.	**Active**
Smartwatch Telemetry Sync	Personnel	1-click-la wearable sensor heart rate and sleep biometrics form-la load panna.	**Active**
1-Click Crisis SOS Beacon	Personnel / Officer	Army 1902 & Tele-MANAS 14416 call seiya + Officer screen-la pulsing red alert kaatta.	**Active**
Cumulative Fatigue Forecaster	Personnel / Officer	7-day sleep debt calculation panni Operational Strain Index (0-100%) and rest alert kaatta.	**Active**
Gamified Morale Badges	Personnel	Continuous self-logging seiyumbodhu daily streak and resilience badges unlock panna.	**Active**
Personnel Monitoring Roster	Officer	Command unit jawans list-ai triage color tags and filters vechu live monitor panna.	**Active**
Soldier Clinical Dossier	Officer	Oru soldier-oda complete historical trends, sleep graph, past notes review seiya.	**Active**
Confidential Counseling Notes	Officer	1-on-1 counseling mudithu findings, treatment plan, next review date secure-ah log panna.	**Active**
Executive Analytics Dashboard	Admin	Army-wide High, Medium, Low triage counts and unit health distribution overview.	**Active**
Command Authorization Master PIN	Admin / Security	Officer/Admin signup prevent panna 4-digit PIN. Admin-aal dynamically rotate seiyya mudiyum.	**Active**
Unit Self-Service Recovery	All Roles	`/forgot_password` moolam Name + Service Unit submit panni password reset seiya.	**Active**
Immutable Activity Audit Trail	Admin / System	All critical events, logins, notes, PIN changes timestamp-udan log aagum.	**Active**
Vernacular Multi-Language	All Roles	Zero-latency-la English, Tamil, Hindi language switch seiya (`localStorage` persistence).	**Active**
Laptop 2-Col & Mobile Drawer UI	All Roles	Zero horizontal overflow, 2-column laptop grid, touch-friendly mobile drawer menu.	**Active**
Vercel Blob Cloud Persistence	Cloud Backend	Serverless cold boots-la SQLite database azhindhu pogamal automatic snapshot sync seiya.	**Active**
PDF Report Generators	Personnel/Officer/ Admin	Soldier Dossier (`/my/export_pdf`), Officer Report, Admin Command Report download panna.	**Active**

Mudivu & System Guarantee: Indha APPS platform ippo military welfare monitoring-kku thevaiyaana anaithu modern tools-udan 100% production-ready aagi Vercel-la live-ah run aagudhu. Soldiers friction-free direct single-step login moolam entry podalaam, officers immediate triage action edukalaam, matrum Senior Admins command overview-ai securely maintain seiyalaam.